

Poloidal magnetic field in the dense plasma focus

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ABSTRACT

Existence of an axial (poloidal) component of magnetic field in the dense plasma focus has been inferred using multiple diagnostics in many laboratories since 1979. It has not received much attention because its origin as well as role in plasma focus physics was unclear till recently. Recent discovery of long-lasting neutron emission perpendicular to the axis in PF-1000 and neutron fluence ratio (end/side) less than unity in Gemini shows that azimuthally accelerated and radially confined deuterons play an observable role in fusion reactions. A spontaneously generated poloidal magnetic field can provide both the azimuthal electric field necessary for acceleration and radial confinement of the ions being accelerated in the acceleration zone. A comprehensive survey of plasma focus research also confirms the role of spontaneously self-organized plasma objects in the fusion reaction process where their three-dimensional magnetic field structure provides a mechanism for accelerating and trapping ions making them repeatedly pass through a dense plasma target. With emerging appreciation of the likely role of the axial magnetic field in plasma focus neutron emission, it becomes imperative to consider models for its origin. This Letter proposes a partial theory of growth of the axial (poloidal) magnetic field via a simple dynamo, with the geomagnetic field as the seed, which converts the kinetic energy of the plasma into energy of the poloidal magnetic field. This theory leads to an experimentally testable proposition.

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From its discovery in the early 1960s,^{1,2} the dense plasma focus (DPF) has always been described and modelled^{3–13} as an axisymmetric z-pinch driven by toroidal magnetic field $\vec{B}_t \equiv B_0 \hat{\theta}$ generated by a poloidal current $\vec{J}_p \equiv J_r \hat{r} + J_z \hat{z}$. Sporadic reports since 1979 inferring existence of an axial component of magnetic field (more correctly described as a poloidal component $\vec{B}_p \equiv B_r \hat{r} + B_z \hat{z}$ created by a toroidal current $\vec{J}_t \equiv J_\theta \hat{\theta}$)¹⁴ using multiple diagnostics in many laboratories¹⁵ have been largely ignored, including by their own authors. This was because there was no clarity concerning its origin and its role in plasma focus phenomenology. This situation has changed recently.

Discovery of long-lasting neutron emission (of ~ 200 ns duration, comparable to the radial Alfvén transit time and much larger than the radial transit time for ~ 100 keV ions) perpendicular to the axis in PF-1000¹⁶ and neutron fluence ratio (end/side) less than unity in Gemini¹⁷ (indicating more neutron emission at 90° than at 0° to the axis) suggests that azimuthally accelerated and radially confined deuterons play an observable role in fusion reactions. A comprehensive review of plasma focus research¹⁸ also suggests that the three-dimensional magnetic structure associated with spontaneously self-organized magneto-plasma objects provides a mechanism for acceleration and trapping of ions,¹⁹ making them repeatedly cross a dense plasma target. This significantly increases the fusion reaction

rate as compared with a thermal plasma of equivalent particle and energy density. Such emerging appreciation of the likely role of the poloidal magnetic field in the fusion process is accompanied by an experimental demonstration^{20–23} of the emission of poloidal magnetic flux from the plasma focus that begins with the start of current and continues well after the pinch disassembly. This complements the difficult and more elaborate magnetic probe measurements of the axial magnetic field by Krauz *et al.*²⁴ on PF-1000.

In spite of these experiments, existence of an axial component of magnetic field in the dense plasma focus is likely to remain an unprovable hypothesis forever. Strictly speaking, an unambiguous measurement of the axial component of magnetic field in a plasma is impossible even in principle. An experiment attempting measurement of one component of a vector field defined at each point in the plasma must simultaneously satisfy many conditions to qualify it as a physically valid measurement.

1. It must demonstrate that the measured numbers are not related to any other component of that vector quantity. In the case of the axial component of magnetic field, the error due to sensitivity of the measurement to the azimuthal or radial components of the magnetic field needs to be quantified. This error estimate is as much as 50% in the magnetic probe measurements.²⁴

2. It must either have a well-defined spatial resolution, deriving experimental numbers attributable to a small region around a well-defined point, or it must have a mathematical transform relationship that enables reconstruction of the spatial distribution of the vector component from many simultaneous measurements. A magnetic probe has²⁴ a well-defined spatial resolution $\sim 1\text{--}2\text{ mm}$. A measurement of the Faraday rotation of the plane of polarization of a laser beam by the *azimuthal* component of the magnetic field can be inverted using an Abel transform. Such transform does not exist for the *axial* component.
3. The measurement process should not alter the quantity being measured. The magnetic probe represents a foreign body immersed in the plasma. It forces the plasma to flow around it, altering the current that is the source of the magnetic field. It also creates a local charge density because of the formation of a Langmuir sheath around the probe body. These processes alter the value of the quantity being measured and produce a spurious contribution to the measurement. These non-quantifiable errors make the very concept of measurement of the axial component of magnetic field at a point within the plasma completely meaningless.

As opposed to measurement, a hypothesis concerning existence of an axial magnetic field is an entirely different proposition. A series of observations of different phenomena which are inexplicable without invoking the hypothesis of existence of an axial magnetic field do not strictly conform to the definition of a scientific proof of such hypothesis—unless this list is sufficiently long to be considered asymptotically equivalent to “a scientific proof.” However, they do suggest that the hypothesis of existence of an axial magnetic field is a valid subject of further research. A list of such observations is described elsewhere¹⁵ and is not repeated here because of space constraints. The latest of such observations is the demonstration^{20–23} of poloidal magnetic flux emission in a plasma focus. Measurement of the returning poloidal magnetic flux outside the plasma can be considered an unambiguous proof of existence of its source, a magnetic moment, which is an indirect proof of existence of an axial magnetic field within the plasma itself. Another example is described as a do-it-yourself experiment at the end of this Letter.

The hypothesis of existence of an axial magnetic field needs to be accompanied by an enquiry into the likely origin of the poloidal magnetic field in an azimuthally symmetric plasma. Such enquiry must begin with the simplest possible mechanism which must be additionally subjected to experimental tests. If and when such tests reveal inadequacy of the suggested model, more nuanced models can be considered. These include a variety of dynamos that convert plasma kinetic energy into magnetic energy,²⁵ paramagnetic drift of charged particles in an imploding plasma,²⁶ constrained dynamics,²⁷ resonant coupling of electron modes,²⁸ and coherent effects in stochastic electrodynamics of plasma.²⁹ It is quite likely that many mechanisms may be operating in tandem. So, lack of agreement between any one model and experiments needs to be treated as an opportunity for a more extensive diagnostic exploration of the pattern of disagreement.

This Letter is intended as a first tentative step on the path of this enquiry. Its scope is, therefore, necessarily and deliberately limited to formulating an experimentally testable proposition concerning the poloidal magnetic field and its role in plasma focus physics rather than

constructing a mathematical formalism that attempts quantitative calculation.

The simplest mechanism for generation of the poloidal magnetic field²¹ is the operation of a simple dynamo seeded by the geomagnetic field where the kinetic energy of the plasma gets converted into the magnetic energy. This is also popularly known as magnetic flux compression or the sweeping along of the poloidal magnetic flux by the moving plasma conductor. Electromotive force induced by the magnetic field in a moving conductor (armature) drives a current in an electrically continuous circuit that creates a magnetic field that can sometimes augment the initial magnetic field.

This can be expressed in terms of the azimuthal component of the generalized Ohm's law, where the electron pressure gradient term does not contribute on account of symmetry,

$$E_\theta = -(\vec{v} \times \vec{B})_\theta + \eta J_\theta. \quad (1)$$

The Hall term is neglected as a model assumption.

The curvature of the plasma focus sheath ensures that the axial component of the geomagnetic field in a frame of reference associated with the device always encounters a radial component of plasma velocity in order to generate an azimuthal component of electric field. Faraday's law would then ensure growth of the axial component of magnetic field at a rate that depends on its magnitude. This growth should be ultimately limited by resistive dissipation of the associated azimuthal current. In this limiting case, the azimuthal current density should reach²¹ a steady value $J_\theta \sim \eta^{-1} E_\theta$. However, the time required to reach this limit may be longer than the duration of the discharge if the resistivity is small. A more basic model would be to consider the zero-resistivity limit of the generalized Ohm's law (1). This is also justified by an estimate²¹ of the magnetic Reynold's number R_m for ball-park numbers for a plasma focus which is much larger than unity.

This heuristically simple model requires a knowledge of the velocity and curvature of the plasma during its propagation. The recently introduced theory of the generalized plasma focus problem,³⁰ which is a conceptual outgrowth of the resistive Gratton Vargas model,^{31–35} can be used to obtain this information. This theory provides a generic description of phenomena similar to a conventional plasma focus, which have plasma formation, plasma propagation, and “approach to symmetry axis” phases. It decomposes the plasma propagation problem into two weakly coupled subproblems by expressing every space and time dependent physical property of the plasma as a product of a scaling parameter (denoted by subscript 0), which contains device-related information, and a scaled variable (denoted by an overtilde), which does not. This transforms the equations governing plasma dynamics (such as Euler equations of fluid dynamics, Maxwell's equations, generalized Ohm's law, and equation of state) into a non-dimensional form, from which all device-related information is removed except that retained in the scaling of material properties related to the equation of state and transport phenomena. These equations include the effects of inertia, acceleration, transport processes, and energy balance.

The scaling parameters for the coordinates, density, magnetic field, and the velocity are, respectively, the anode radius “a,” the mass density of the fill gas ρ_0 ,

$$B_0 = \frac{\mu_0 I(t)}{2\pi a \tilde{r}} \quad \text{and} \quad v_0 \equiv \frac{B_0}{\sqrt{2\mu_0 \rho_0}}. \quad (2)$$

The scaled time τ is related to the charge flow^{30–35} as

$$\tau = Q_m^{-1} \int I(t) dt; \quad Q_m = \pi \mu_0^{-1} a^2 \sqrt{2\mu_0 \rho_0}. \quad (3)$$

The first subproblem determines the evolution of an imaginary traveling surface of rotation $\psi(\tau, \tilde{r}, \tilde{z}) = 0$, moving with a perpendicular local velocity that equals the scaling parameter for velocity. This surface, called the Gratton–Vargas (GV) surface, obeys the Gratton–Vargas (GV) equation.^{30–35} Solutions of the GV equation are defined to within an arbitrary multiplier, which can be chosen to provide a convenient normalization.³⁰ They form a two-parameter family of surfaces $\psi(\tau, \tilde{r}, \tilde{z}) = \psi_0$ with the parameter τ representing the time and the parameter ψ_0 representing a continuum of similar surfaces in the neighborhood of the principal surface $\psi(\tau, \tilde{r}, \tilde{z}) = 0$. The solution has the following form:

$$\psi(\tau, \tilde{r}, \tilde{z}) = \left(2N\tilde{z} - s \left(\tilde{r} \sqrt{\tilde{r}^2 - N^2} - N^2 \text{Arc Cosh} \left(\frac{\tilde{r}}{N} \right) \right) - \tau \right) - \psi_{GV} = \psi_0. \quad (4)$$

The number N is an invariant, determined by boundary conditions (algebraically defined shapes of the anode and insulator), along every characteristic curve that start from a point on the anode or insulator and is perpendicular to the GV surface (4) at each instant τ .

Solutions of the GV equation for a Mather type plasma focus are known³⁶ to resemble the experimentally determined shapes of the plasma in various phases of the discharge. A numerical procedure to determine this surface for a given device is available.^{30–35} This includes the determination of the current in the device as a function of time.

The second subproblem consists of determination of evolution of the scaled variables in the dimensionless space and time. This consists of solutions of the non-dimensional equations governing the plasma dynamics (which could be any variety of fluid model). These solutions must be universally valid for all the devices that conform to the definition of the generic problem, irrespective of device parameters, provided the scaled material properties have identical values.

The spatial distributions of all the scaled plasma variables should move at the scaling velocity along with the GV surface, by definition. The plasma armature can then be described with respect to a moving spatial region $\Delta\psi(\tau)$ whose front and rear boundaries are the two surfaces $\psi(\tau, \tilde{r}, \tilde{z}) = 0$ and $\psi(\tau, \tilde{r}, \tilde{z}) = \delta\psi_0$, where $\delta\psi_0$ is a number $\sim O(R_m^{-1})$,

$$\vec{v} = v_0 \hat{\xi} \quad \text{within } \Delta\psi, \\ = 0 \quad \text{otherwise,} \quad (5)$$

where $\hat{\xi}$ is a unit vector normal to the GV surface,¹⁵ $\hat{\xi} = (\tilde{r}s\tilde{r}^{-1} \sqrt{\tilde{r}^2 - N^2} + N\tilde{r}^{-1}\tilde{z})$, and s is the sign of the radial velocity:

The zero-resistivity limit of the generalized Ohm's law then gives the following relation, valid only in the region $\Delta\psi(\tau)$, using the vector potential $\vec{A}$,

$$E_0 = -v_0 (\hat{\xi} \times (B_r \hat{r} + B_z \hat{z})) \cdot \hat{\theta} \\ = -v_0 (B_r (N\tilde{r}^{-1}) - s\tilde{r}^{-1} \sqrt{\tilde{r}^2 - N^2} B_z) \\ \Rightarrow -\frac{\partial A_0}{\partial t} = - \left(\frac{1}{\sqrt{2\mu_0 \rho_0}} \frac{\mu_0 I(t)}{2\pi a \tilde{r}} \right) \left(-\frac{\partial A_0}{\partial \tilde{z}} (N\tilde{r}^{-1}) \right. \\ \left. - s\tilde{r}^{-1} \sqrt{\tilde{r}^2 - N^2} \frac{1}{a \tilde{r}} \frac{\partial}{\partial \tilde{r}} (a \tilde{r} A_0) \right). \quad (6)$$

Introducing the flux function

$$\Phi \equiv \int_0^{\tilde{r}} B_z 2\pi r dr = \int_0^{\tilde{r}} \left(\frac{1}{r} \frac{\partial r A_0}{\partial r} \right) 2\pi r dr = 2\pi r A_0 = 2\pi a \tilde{r} A_0 \quad (7)$$

and using the GPF scaling (2) and (3), Eq. (6) can be written as

$$\frac{\partial \Phi}{\partial \tau} = -\frac{N}{2\tilde{r}} \frac{\partial \Phi}{\partial \tilde{z}} - \frac{s}{2\tilde{r}} \sqrt{\tilde{r}^2 - N^2} \frac{\partial \Phi}{\partial \tilde{r}}. \quad (8)$$

Equation (8) can be written in the Jacobi form³⁷ $p_\tau + H = 0$, where $p_\tau \equiv \partial_\tau \Phi$, $p_{\tilde{r}} \equiv \partial_{\tilde{r}} \Phi$, $p_{\tilde{z}} \equiv \partial_{\tilde{z}} \Phi$, and

$$H = \frac{s}{2\tilde{r}} \sqrt{\tilde{r}^2 - N^2} p_{\tilde{r}} + \frac{N}{2\tilde{r}} p_{\tilde{z}}. \quad (9)$$

Obtaining a general solution of this equation is beyond the scope of this Letter and will be presented elsewhere. Some observations are however immediately accessible.

- Since the Hamiltonian (9) has no explicit time dependence, it is conserved along every characteristic curve³⁰ that obeys the associated Hamilton–Jacobi equations.^{30,37}
- Since the Hamiltonian has no explicit dependence on the axial coordinate, $p_{\tilde{z}} \equiv \partial_{\tilde{z}} \Phi$ is conserved on each characteristic curve.
- Because of the explicit dependence of H on the radial coordinate, $p_{\tilde{r}} \equiv \partial_{\tilde{r}} \Phi$ evolves with time. This evolution depends on the constant value of $p_{\tilde{z}}$ which must be governed by boundary conditions.

These observations in turn imply the following:

The electric field in the region $\Delta\psi(\tau)$ is

$$E_0 = -\frac{\partial A_0}{\partial t} = -\frac{\mu_0 I(t)}{\pi a^2 \tilde{r} \sqrt{2\mu_0 \rho_0}} \frac{\partial \Phi}{\partial \tau} = \frac{\mu_0 I(t)}{\pi a^2 \tilde{r} \sqrt{2\mu_0 \rho_0}} H. \quad (10)$$

The axial magnetic field is

$$B_z = \frac{1}{a \tilde{r}} \frac{\partial}{\partial \tilde{r}} (a \tilde{r} A_0) = \frac{1}{2\pi a^2 \tilde{r}} \frac{\partial \Phi}{\partial \tilde{r}} = \frac{p_{\tilde{r}}}{2\pi a^2 \tilde{r}}. \quad (11)$$

The radial magnetic field is

$$B_r = -\frac{\partial A_0}{\partial \tilde{z}} = -\frac{1}{2\pi a^2 \tilde{r}} \frac{\partial \Phi}{\partial \tilde{z}} = -\frac{p_{\tilde{z}}}{2\pi a^2 \tilde{r}}. \quad (12)$$

The current density in the region $\Delta\psi(\tau)$ is

$$J_0 = -\mu_0^{-1} \left(\frac{\partial B_r}{\partial z} - \frac{\partial B_z}{\partial r} \right) = -(2\pi a^2 \mu_0)^{-1} \frac{\partial}{\partial \tilde{r}} (\tilde{r}^{-1} p_{\tilde{r}}). \quad (13)$$

A complete solution of the problem would involve calculating vacuum fields outside the region $\Delta\psi(\tau)$, matching fields, and their derivatives at all points on the boundaries of the region at all times and applying initial and boundary conditions. This is beyond the scope of this Letter.

However, this partial exercise does bring out the following significant points:

1. Evolution of the axial magnetic field during plasma propagation is clearly seen as a property of the simple dynamo. In the absence of an initial axial magnetic field, the Hamiltonian (9) has the initial value zero on every characteristic curve and its invariance

- ensures that it remains zero. On the other hand, in the presence of even a minimal axial magnetic field, say of geomagnetic origin, H is clearly nonzero and the magnetic flux does change with time via (8) as does its spatial distribution, via (11).
- Equation (8) can be solved using the method of separation of variables, which gives an exponential dependence of Φ on τ involving the separation constant which must be determined using boundary conditions. An exponential growth of poloidal magnetic field is possible if the separation constant has an appropriate positive value. This boundary value problem is yet to be fully solved.
 - The boundaries of the region $\Delta\psi(\tau)$ are governed by the magnetic Reynold's number R_m , which would be affected by introduction of impurities in the plasma during the discharge. The fact that an increasing poloidal magnetic flux emission signal is detected outside the cathode^{20–23} between the start of current and current derivative singularity in some, but not all, shots suggests that boundary conditions on p_z do play a determining role.
 - The energy required for the growth of poloidal magnetic field must be derived from the power source for the plasma focus.³⁸ The contribution to the total power $W(\tau) = \int d^3\vec{r} \vec{J} \cdot \vec{E}$ from this process would be proportional to integration over the region $\Delta\psi(\tau)$ of the product of J_θ given by (13) and E_θ given by (10). The above-mentioned exercise shows that E_θ is proportional to the current $I(\tau)$ that drives the plasma via the velocity v_0 . But the current density J_θ has no relation with $I(\tau)$ since it depends on the spatial structure of the flux function Φ . Therefore, the equivalent loop voltage in the circuit element representation of the plasma focus³⁸ $\sim W(\tau)/I(\tau)$ would involve a term proportional to the geomagnetic field but independent of the charging voltage.
 - The current that enters the device from the pulse power source must equal the total current inside the device which should be

$$I_{\text{dev}} = \int \vec{J} \cdot d\vec{S} = \int J_r \hat{r} \cdot d\vec{S} + \int J_\theta \hat{\theta} \cdot d\vec{S} + \int J_z \hat{z} \cdot d\vec{S},$$

$$= \int J_r r d\theta dz + \int J_\theta r dr dz + \int J_z r dr d\theta. \quad (14)$$

However, when the current inside the plasma focus is “measured” using magnetic probes looking at the azimuthal magnetic field that results from the poloidal current $\vec{J}_p \equiv J_r \hat{r} + J_z \hat{z}$ a tacit assumption is made that $J_\theta \approx 0$.

This should explain why even sophisticated MHD codes, which neglect this dynamo effect, overestimate the observed current.^{39–41} The deficit may simply be the current that circulates in the azimuthal direction in the actual experiment. An artificial current loss factor that has to be incorporated in the Lee model⁴² to bring its calculated waveform into coincidence with an experimental waveform may also be partly related to neglect of the dynamo. This mechanism might explain the inference of an apparent current loss from interpretation of measurements⁴³ that is based on the assumption that the magnetic field is purely toroidal.

This outline of a partial theory of generation of the poloidal magnetic field, based on a simple dynamo seeded by the local geomagnetic field and driven by a curved plasma armature, suggests the following simple experimental test. It consists of creating a uniform axial magnetic field over the entire volume of a small plasma focus using a large

Helmholtz coil powered by a DC source of variable polarity and a maximum amplitude not more than two times that of the local geomagnetic field. By monitoring the response of the plasma focus using the current derivative signal, its integration and the poloidal magnetic flux emission signals^{20–23} to the polarity and magnitude of the externally produced axial magnetic flux, it should be possible to test this model. A Lee model⁴² fit to the measured current waveform should then show a distinct variation of the radial current fraction parameter with the external magnetic field near a value that nullifies the local geomagnetic field. Note that a non-uniform high magnetic field produced by a permanent magnet or an electromagnet placed somewhere in the vicinity of the plasma focus will not correspond to the model and will have no interpretive value. Also, excessively high value of applied magnetic field would disturb the plasma armature, leading to observations of reduced interpretive value.

The significance of such experiments goes beyond the mere validation or refutation of a model. In case the apparent current loss does vary by introduction of a small amount of external poloidal magnetic flux, it potentially provides a means of external control of the properties of a plasma focus. For example, would the neutron yield in Gemini be increased by nullifying the ambient geomagnetic field? Would the neutron fluence anisotropy, reported to be less than unity,¹⁷ become different? On the other hand, if the apparent current loss and the poloidal magnetic flux emission signals^{20–23} show no dependence on the externally created flux, does this imply the existence of a self-excited autonomous dynamo?

Reservations about committing time and resources for pursuing experimental research outlined above are naturally to be expected in view of several decades worth of silence on this topic in plasma focus literature. However, minimal efforts are sufficient to replicate the Nikulin effect,⁴⁴ which should dramatically address such reservations. Dr. Valery Nikulin of the Lebedev Institute of the Russian Academy of Sciences placed a cone-shaped copper foil (see Fig. 1) on the anode of



FIG. 1. The Nikulin effect: On the right is a cone-shaped copper foil similar to the one placed on the plasma focus anode. On the left is the cone retrieved after a shot. The cone is twisted, not radially collapsed, showing the presence of a torque. Reproduced with permission from Nikulin, private communication (2005). Copyright 2005, Dr. Valery Nikulin.⁴⁴

his 2.5 kJ plasma focus⁴⁵ in 2005. After the shot, the cone was not radially imploded but twisted, showing the presence of a torque. A purely azimuthal magnetic field, as assumed traditionally,^{3–13} cannot produce a torque, demonstrating that a poloidal magnetic field must exist.

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AUTHOR DECLARATIONS

Conflict of Interest

The author has no conflicts to disclose.

Author Contributions

S. K. H. Auluck: Conceptualization (lead); Formal analysis (lead); Investigation (lead); Methodology (lead); Software (lead); Visualization (lead); Writing – original draft (lead).

DATA AVAILABILITY

Data sharing is not applicable to this article as no new data were created or analyzed in this study.

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